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(1) Extension Program/ Project Title: Building Sustainable Livelihood: CAPSU Burias i-ProTech for Dumolog Dried Fish Association, Roxas City	
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(2) Category (please check) ☐ Education and Humanities ☐ Agriculture/Fisheries/Livelihood ☐ Health/Environment ☒ Engineering and Technology

(4) Background (WHAT, WHERE is the proposed project, **WHO** will benefit, what **PRODUCT/SERVICE** will address the identified needs/resources. Introduce the product/service that will address the identified needs and resources (general statement). Introduce the target group, organization/community to be served, and the project beneficiaries briefly. Include related literature consistent with the subject – to make the proposal scholarly)

The “**Building Sustainable Livelihood: CAPSU Burias i-ProTech for Dumolog Dried Fish Association, Roxas City**” project is a technological advancement initiative aimed for empowering sustainable fisheries by introducing CAPSU Burias developed technologies. These innovations focus on providing a more efficient, convenient and environmentally friendly method of drying fish- the **Ricehull-Fueled Fish Dryer** and **Ricehull Charcoal Briquetting Machine**.

The fish dryer utilizes rice hulls as fuel, tapping into the abundant availability of this agricultural waste. To enhance productivity, the by-product of the combustion process carbonized rice hull, is further processed into charcoal using of a charcoal briquetting machine.

The primary beneficiaries of this project are the fisherfolks, dried fish processors and sellers associated with the **Dumolog Dried Fish Association**, located in Brgy. Dumolog, Roxas City. The collaboration with the Local Government will be sought to ensure the sustainability and successful implementation of the project.

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According to Doe and Olley (1990), drying is one of the earliest known means of preserving foods. Fish have been dried by traditional methods everywhere fish are caught. By far the most widely used method of drying fish is direct exposure to sunlight. Though, sun drying is most simple and economical method of fish preservation. Sun drying still has many limitations, such as long period of drying during cloudy climate. Oftentimes, sun drying of fish results in low quality as a result of slow drying, insect infestation and contamination from air borne dust (Curran CA, Trim DS, 1985). For getting better quality-dried fish, it is very essential to use improved methods of fish drying. Moreover, it is also important to maintain required hygiene during the different phases of fish drying (Relekar et al., 2013).

Rice hulls are recognized as one of the major potential sources of energy production (Sivakumar et al., 2010). Rice hull is known to be an important source of energy. Rice hulls biomass materials maintain net calorific values between 6000 kcal/kg and 3000 kcal/kg; this is almost half that of natural gas and fossil fuel. Moreover, rice hull can be used as an admixture to the composition of fuel briquettes with more flammable materials (Romallosa & Kraft, 2017). Rice hull charcoal can be mixed with other organic materials to produce organic fertilizers to replace chemical fertilizers, reducing production costs for farmers in the current crisis (Homchat & Ramphueiphad, 2022). Other uses of rice hull charcoal may include Growing Media for plant (Setyawan, 2023), deodorization, water purification, and soil conditioning (Koyanagi et al., 2019).

5) RATIONALE (WHY should the project be implemented? **WHAT need** is the project trying to address? – how the problem is affecting the targeted beneficiaries, previous efforts done and status, any legal bases and related literature supporting the proposed project, link the relevance of the project to development priorities like mandate and development goals, the comparative advantage to handle the project, and **WHY** act now and not later)

CAPSU, through its Agricultural and Biosystems Engineering program, recognizes the immense potential for agri-aqua industrial development in the province and the entire Western Visayas Region 6. One of the program objectives of the Department of Agricultural Engineering, CAPSU Buriás Campus is to effectively transfer developed and matured agricultural engineering technologies, along with knowledge acquired through research, to the country side. This strategic initiative is designed to enhance agricultural productivity and profitability.

The extension banner program of the Department, the Barangay Unification towards Innovative Learning and Development (BUILD) program is guided by the principle that national development should start from the community, at the grass root level (families) and towards the advancement at the barangay level.

DABE has developed a rice hull fueled fish dryer that can efficiently and conveniently dry fish. The technology provides a controlled drying environment, allowing precise regulation of temperature and humidity ensuring a consistent drying conditions as compared to the variable weather conditions associated with sun drying. Moreover, the machine can operate consistently throughout the year, enabling continuous production and reduces the impact of seasonal variations on drying efficiency. The controlled drying conditions prevent contamination from external factors, such as dust, insects and pollutants, leading to higher-quality fish products.

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Moreover, as the machines uses rice hull which can be sourced locally, it reduces the dependence on traditional energy sources, making the drying process more energy-efficient and sustainable, contributing to environmental conservations efforts.

To address the challenge of disposing the combustion by-product, the carbonized rice hull, an intervention will be provided with the briquetting machine capable of converting the carbonized rice hull into charcoal. This finished product could be used as fuel, growth media for plants, and filter.

Hence, the drying process becomes more efficient, reliable, and environmentally friendly, offering a sustainable alternative to traditional sun drying methods.

(6) GOALS AND OBJECTIVES (what do you **WANT TO ACHIEVE**? This presents what the proponent hopes to accomplish through the project and what can address the cause of the core problem. Captures changes in people's lives at the societal level – the **IMPACT**. The **BIG PICTURE**.)

1. To promote and facilitate transfer of technology developed through research.
2. To disseminate CAPSU Burias mature technologies - fish dryer, to fisherfolks and end users.
3. To train local fisherfolks and dried fish processor in drying fish utilizing the developed fish dryer.
4. To provide technical and consultancy services to the technology adaptor – Dumolog Dried Fish Association in (a) operation and maintenance of the fish dryer, and (b) financial literacy and entrepreneurial skills management.

(7) PROJECT DESCRIPTION (what are the **features** of the project process/tools used in the project ID, location, duration, beneficiaries, results chain/framework, project cycle – [*needs/gaps identified, set of solutions noted to address the needs*])

The “**BUILD: CAPSU Burias Agri-Aqua ProTech for Empowering Sustainable Fisheries**” project is designed to transfer developed technology for a more **efficient** and **convenient** method of **drying fish using ricehull**, and converting combustion by-product into ricehull charcoal thus promote **entrepreneurship** and **empower** the Dumolog Dried Fish Association by enhancing productivity, ensuring high production, and delivering quality products.

(8) PROJECT IMPLEMENTATION AND Monitoring & Evaluation SYSTEM (how will the project be implemented, how will the project be tested if it is working, when - detailed activities to be undertaken by component to generate the outputs, support the description of activities through a matrix or work plan identifying how each activity will be delivered, who will be responsible for **WHAT, WHEN, WHERE** – state your Activities, Work Plan - Project Schedule supported with a Gantt chart, Project Management [proponent, partners and their roles and responsibilities including counter parting scheme, if any, capacity to handle the project, and accompany with the simple management structure. Provide theoretical background and apply it to the project. Describe how the project will be monitored and evaluated during the planning/pre-implementation, implementation, and operation/post-implementation phases. M&E framework must be accompanied by a **DATA Collection Plan**)

1. Profiling and Benchmarking

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The profile of the Dumolog Dried Fish Association members will be gathered using the prescribed forms from the Extension Office. Profiling and benchmarking is conducted to identify the demographic and economic status of the beneficiaries.

2. Collaboration with Target Agencies and Stakeholders

The success of the project will be dependent on the effective partnership with other agencies and stakeholders.

CapSU Burias Extension Department will be the implementing agency that will spearhead in the conduct of techno-demonstrations and other extension-related activities in order to meet the needs and demands of the people in the community.

CAPSU RDE will provide the developed technologies for dissemination.

Roxas City LGU will serve as the data provider in identifying the target beneficiaries of the project. They will also serve as partner agency in the establishment of the project and in the conduct of trainings and other extension-related activities.

3. Orientation and Training Need Assessment

In this activity, the identified beneficiaries will be oriented on their roles and responsibilities towards the implementation of the project. The priority needs will be identified through the conduct of training needs assessment.

4. Launching and MOA signing

This is the presentation and unveiling of the project. Memorandum of agreement between the implementers, collaborating agencies, stakeholders, and beneficiaries is conducted. The roles and responsibilities of the parties will also be stipulated and emphasized during the MOA signing.

5. Implementation

- Conduct of techno-demo on the operation of the fish dryer and charcoal briquetting machine and other extension-related activities are executed as planned.
- Provide technical and consultancy services in the operation and maintenance of the machines.
- Provide trainings on financial literacy and entrepreneurial skills management.

6. Project Monitoring and evaluation

Quarterly and semi-annual monitoring and evaluation of the socio-economic effect of the use of the transferred technologies will be conducted by the implementers and stakeholders to identify the strengths and weaknesses of the project. Problems encountered and the lessons learned will be identified.

7. Impact assessment

Assessment in terms of research study will be conducted to measure the sustainability and effectiveness of the implemented project.

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(9) Work Plan (how will you achieve your objectives?)

Activity	2024											
	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec	Jan
Profiling/Benchmarking												
Collaborate with target agencies and stakeholders												
Orientation and training needs assessment survey												
Launching and MOA signing												
Implementation of the program												
Project Monitoring and evaluation (2024)												
Impact Assessment (after 1 year and 3 years of implementation)												

10) Budget (how much will it cost)

Activities	Materials/Foods	Fuel
Orientation and TNA	3,500.00	1000.00
Launching of the project	10,000.00	500.00
Implementation (trainings)	10,000.00	4,000.00
Project monitoring	10,000.00	4,000.00
Impact Study	5,000.00	2,000.00
Total	38,500.00	11,500.00
Total budgetary requirement	50,000.00	